

MARCUS BELL

Innovation Lead · Principal Product Owner · Principal Architect

Digital Product Evolution · AI & Robotics · IoT & Cloud Platforms

Niederkasseler Kirchweg 122, 40547 Düsseldorf | +49 176 22940533 | marcus.bell@web.de



PROFESSIONAL EXPERIENCE

Vorwerk Group, Düsseldorf

Innovation Lead / Principal Product Owner / Principal Architect | 2017 – present

End-to-end product and architecture ownership for digital ecosystems, platform services and AI-based value-added services within the Vorwerk Group – from product vision, roadmap and backlog through to operational management.

SEVEN PRINCIPLES AG (7P Solutions & Consulting), Ratingen

Management Consultant / Team Manager Enterprise Architecture | 2009 – 2017

Strategic consulting and technical solution ownership in the areas of digitalization, IoT and cloud transformation. C-level advisory, business development and establishment of the enterprise architecture team.

Valtech Deutschland GmbH, Düsseldorf

Senior Consultant – Telecommunications & Telematics | 2002 – 2009

Enterprise architecture, solution architecture and integration management for large-scale telecommunications platforms – focus on Vodafone, BMW and international clients.

Previous Positions

ICL Deutschland GmbH, Düsseldorf – Senior Consultant eBusiness | 2001

Quark Deutschland GmbH, Ludwigsburg – Software Architect | 1997 – 2001

KW Software GmbH, Lemgo – Development Engineer / Software Architect | 1995 – 1997

PROJECT REFERENCES

Vorwerk Group (2017 – present)

01/2024 – now | Innovation Lead AI Services | Vorwerk Elektrowerke

Design and implementation leadership for AI-based value-added services leveraging ML and AI technologies: LLM integration, computer vision, structure recognition, object detection, agent systems, humanoid robotic systems, semantic and spatial world models.

01/2023 – 12/2023 | Principal Product Owner Robotic Clouds | Vorwerk Elektrowerke

Robot cloud migration, vendor management and change management. Stakeholder management and technical leadership of teams. Corporate change and organizational transformation.

11/2017 – 12/2022 | Principal Architect Cleaning Ecosystem | Vorwerk Services

Vision, conceptualization and development of the digital ecosystem for cleaning robots and cleaning appliances: robot fleet management, cloud system architecture, apps and customer touchpoints. Technical leadership of teams, stakeholder management and partner management.

SEVEN PRINCIPLES AG (2009 – 2017)

07/2012 – 11/2017 | Unit Manager Enterprise Architecture | Seven Principles AG

Unit Manager Enterprise Architecture: responsibility for domain portfolios, team setups, CIO advisory, architecture management, portfolio management, Internet of Things and digitalization.

06/2016 – 11/2017 | Digital Architect – Digital Transformation | BEKO Technologies

C-level advisory on digital transformation: formulation of vision and mission, focus and objectives. Service design thinking workshops and business model canvas. Development of business scenarios, definition of business capabilities and gap analysis. IT roadmap, initiation of measures and ongoing advisory throughout the transformation program.

12/2016 – 04/2017 | IoT Architect – Harvester Predictive Maintenance | Holmer Maschinenbau GmbH

Solution design for harvester predictive maintenance. Deployment of Microsoft IoT Azure PdM, Power BI, Huawei IoT Platform and IoT Gateway AR 500 on the Open Telekom Cloud. Device onboarding (CAN bus, NB-LTE), IoT platform configuration, IoT agent installation and implementation of a custom software application.

09/2016 – 12/2016 | IoT Architect – Multi-Cloud IoT Platform | T-Systems

Platform architecture and security architecture for the Azure platform within the T-Systems IoT multi-cloud landscape. Identity & access management, role model, provisioning of user and access rights for service access.

06/2016 – 09/2016 | IoT Architect – IoT Showcase Innovation Lab | HUAWEI

Vision, concept and development of an IoT showcase for the HUAWEI Innovation Lab: smart tracking and smart parking. Device onboarding with gateway and sensors (HUAWEI UBlox/SARA), connectivity (NB-LTE, WiFi, GPRS) and platform architecture on AWS Cloud (Lambda, DynamoDB).

01/2016 – 06/2016 | Agile Architect – Mobility Platform | Union Investment Group

Architecture concept and evaluation of enterprise mobility platforms for a mobile app factory. Backend integration architecture. Concept for the agile operations and delivery model in the service lifecycle.

01/2015 – 10/2015 | Enterprise IoT Architect – IoT Portfolio Evolution | TÜV Rheinland Gruppe

CIO advisory on service portfolio evolution in the context of digital healthcare. Use cases for wearables in occupational health services. Service design thinking workshops, feasibility study and service portfolio evolution.

06/2014 – 10/2014 | Enterprise Architect – BSS Evolution Roadmap | Telefónica Deutschland

Evolution roadmap for the BSS domain (Business Support Systems, CRM) in the context of the E-Plus and Telefónica merger aligned with IT strategy: digital telecommunications products, hybrid products, real-time provisioning, wholesale strategy and integration concept.

05/2012 – 04/2014 | Solution Architect – API Management | Vodafone Deutschland

Design and feasibility assessment across multiple projects for various telecommunications products. Coordination within projects, partner management and further development of the integration domain. Projects (selection): Mobile Wallet, Near Field Communication (NFC), retail shop integration, provisioning of M2M/IoT devices (wearables).

07/2010 – 11/2012 | Enterprise Architect – Cross Domain Integration Blueprint | Vodafone D2

Development of an architecture vision and domain blueprint for the target architecture of the integration domain. Definition of system capabilities, requirements workshops and vendor selection. Presentation of results to the architecture boards.

09/2009 – 11/2015 | Enterprise Architect – Integration Gateway Platform Evolution | Vodafone D2

Development of the target architecture and guidance of platform evolution. Service governance and integration strategy concepts. Requirements workshops with business units, operational departments and architecture boards. Engineering support as lead architect. Vendor SPOC for Chinese and Indian development teams.

05 – 09/2009 | Enterprise Architect – BSS Domain Architecture Blueprint | VRS Luxemburg

Concept and blueprint for the BSS systems architecture. Process analysis and optimization with business units. Requirements workshops and development of the domain architecture.

Valtech Deutschland / Vodafone (2002 – 2009)

2008 – 2009 | Lead Architect – ConnectedDrive Infotainment Platform | BMW / ATX

Development of the solution architecture for the infotainment platform within the ConnectedDrive program. Design of the architecture for generic integration of partner services (services, content providers) and creation of the AppStore integration concept.

2007 – 2008 | Lead Architect – SOHO/SME Contract Renewal Portal | Vodafone NL

Architecture and high-level design for a contract renewal portal for business customers. Organization and facilitation of requirements workshops with business and IT departments. Implementation oversight and definition of acceptance criteria.

2006 – 2007 | Architect – POS Reference Data Optimization | Vodafone D2

Business analysis of reference data management for POS applications. Optimization of the data model and reference data structures. Resolution of performance bottlenecks in offline tariff data systems.

2005 – 2006 | Integration Manager – Retail Application for Shops | Vodafone D2

Integration management for a central POS application. Responsible for migration to a new product: escalation management, engineering team support and collaboration with offshore development departments.

2004 – 2005 | Architect – Common User Management | Vodafone D2

Development of a generic user management system for POS applications with high security standards and a multi-tenant, role-based access concept for shops and partner shops.

2003 – 2004 | Integration Manager – Specialized Retail Online | Vodafone D2

Responsible for the quality of software releases: definition of acceptance criteria, test automation and introduction of continuous integration.

Earlier Project References

2001 – 2002 | Architect – User Care Frontend | ePlus / KPN

Design and implementation of a user care frontend for call center agents to access customer data. Multi-tenant frontend concept with role-based access control.

1997 – 2001 | Software Architect – eBusiness Platform Mercury | Quark Deutschland GmbH

Design and implementation of modules for an eBusiness platform (customer and billing system, CRM). Responsible for a statistics module based on OLAP and the development of a reliable messaging pattern in the system backend. Management of offshore teams in India.

1995 – 1997 | Software Engineer – IDE for PLC Controllers | KW Software GmbH

Design and implementation of a generic code generator within a complex compiler framework. Foundation for a series of code generators. Development of an algorithm for generating complex data structures (high-level languages) on controllers (binary code).

PATENTS

Self-propelled Floor Processing Device

Development of an evaluation unit that automatically detects and registers restricted zones (no-go areas) in an environment map based on behavioral parameters and movement paths, or adjusts existing zones.

Floor Cleaning Device with Floor Detection Method

A system for cleaning floor surfaces with a control unit and electric motor. The technology utilizes induction voltages and return currents between stator and rotor to analyze the floor surface or to control movement sequences more precisely.

EDUCATION

Diploma in Mathematics (Dipl. Math.) | University of Bielefeld | 1989 – 1995

CERTIFICATIONS AND PROFESSIONAL DEVELOPMENT

- TOGAF Foundation
- Scrum Product Owner
- Professional Scrum Master
- PRINCE II Foundation Level
- ITIL / Service Management
- SAlFe (Scaled Agile Framework)
- Professional Requirements Engineering
- Leadership Development 7P
- Winning Complex Sales
- Tibco Product Bootcamp
- Unified Process Distilled Workshop
- Design Patterns Training
- Soft Skills Seminar “Cicero”
- Rational Rose Training Certificate
- XML and Java Workshop
- Object Oriented Design with C++

PUBLICATIONS AND CONFERENCES

- Objekt Spektrum Online (2008): “Flexible Solutions for Order Fulfilment in Telecommunications Using SOA Concepts”
- SOA Congress Mainz (2007): “Flexible Solutions for Order Fulfilment in Telecommunications Using SOA Concepts”

- OOP Munich Vendor Track (2006): “Business Process Management in Provisioning and Order Fulfilment”
- Valtech Whitepaper (2006): “Order Fulfilment and Provisioning Solutions”

AVAILABLE REFERENCES

- General Higher Education Entrance Qualification (Abitur Certificate)
- Diploma Examination Certificate and Diploma (Diploma in Mathematics)
- Employer References: KW Software GmbH, Quark Deutschland GmbH, ICL Deutschland, Valtech Deutschland GmbH